TOOLS FOR VARIANT VISUALIZATION AND ANALYSIS AT THE RAT GENOME DATABASE AND THE ALLIANCE OF GENOME RESOURCES

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https://rgd.mcw.edu

ALLIANCE of GENOME RESOURCES FOUNDING MEMBER

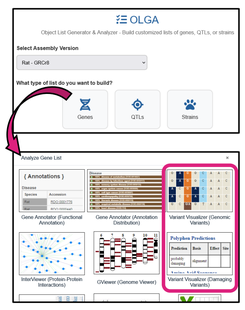
GLOBAL CORE BIODATA RESOURCE

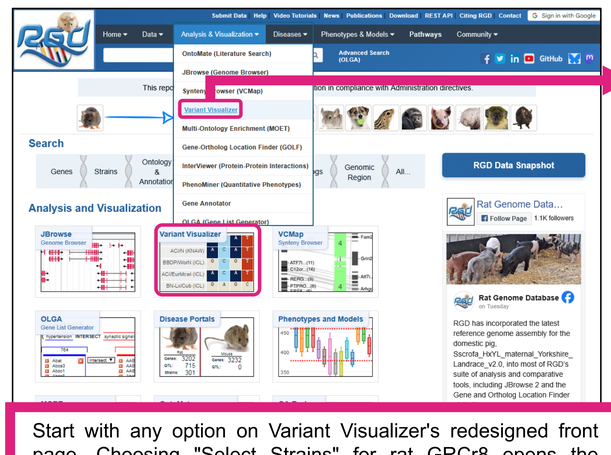
Abstract:

The Rat Genome Database (RGD, https://rgd.mcw.edu) is a cross-species knowledgebase with genomic, phenotypic and disease data for rat (*Rattus norvegicus*) and ten other mammalian species. Likewise, the Alliance of Genome Resources knowledgebase (the Alliance, https://www.alliancegenome.org) integrates data for nine phylogenetically divergent species from human to yeast into a single resource for comparative genomic analyses. Both resources leverage AI to improve existing software and develop new tools for finding and analyzing variant data across species. RGD has a substantial corpus of variant data from the Hybrid Rat Discovery Panel (HRDP) and the European Variation Archive (EVA) for GRCr8 and previous rat assemblies, as well as variants for human, mouse, dog, pig and vervet. These data can easily be explored using RGD's Variant Visualizer tool which can be accessed directly or from RGD's updated search tools such as the Genome Viewer (GViewer) ontology-based search and the Object List Generator & Analyzer (OLGA). Recent updates to the Variant Visualizer tool make the search and filter processes more intuitive. Once options for strain/sample and either a genomic region or gene list have been selected, users can either explore the unfiltered set of variants that meet their search criteria or apply filters, including limiting results to non-synonymous or possibly/probably damaging variants. The result pages are more flexible, with improved navigation and new functionality to allow users to rearrange and group rows to highlight similarities in variant patterns. The Alliance Protein Annotation and Variant Inspector (PAVI) tool, released as a prototype, is a novel analysis tool for both wild type proteins across species and altered proteins with known variants embedded in them. The user selects genes and transcripts across any of the nine Alliance species plus, if desired, one or more variants for each. The tool leverages the genomic nucleotide sequence, transcript definitions, and variant genomic positions to produce amino acid translations. The resulting protein sequences with and without embedded variant(s) are then aligned in real time and presented to the user. The results page displays information about the selected genes, transcripts, and variants; the alignment statistics; and the alignment itself, with variant amino acids highlighted to help users easily explore the effects of their selected variants on the corresponding protein sequences.

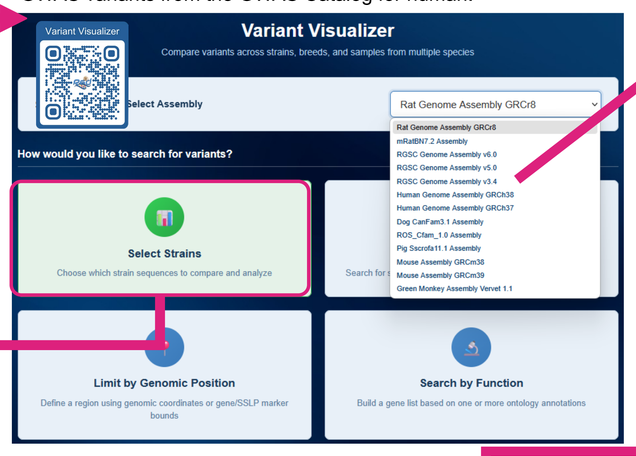
RGD: Variant Visualizer

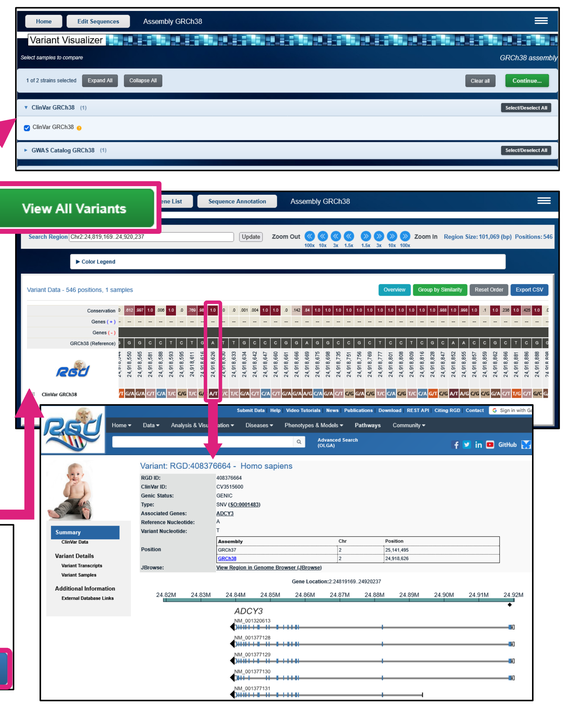
To start, access Variant Visualizer from the RGD "Analysis & Visualization" menu, the tile on the home page or the gene list toolbox in tools like OLGA.



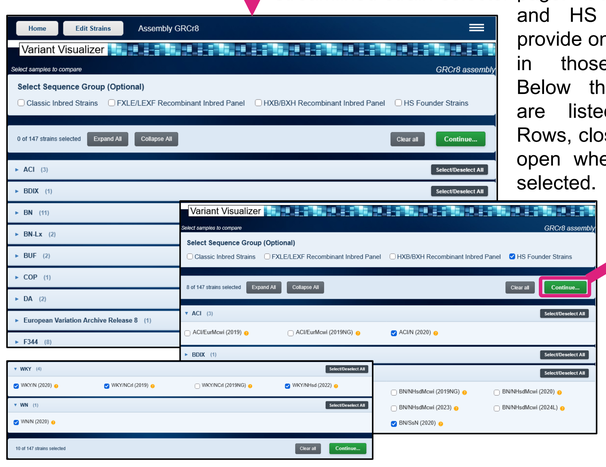


RGD has variants for multiple species and multiple assemblies. For human, these include clinically relevant variants from ClinVar and GWAS variants from the GWAS Catalog for human.

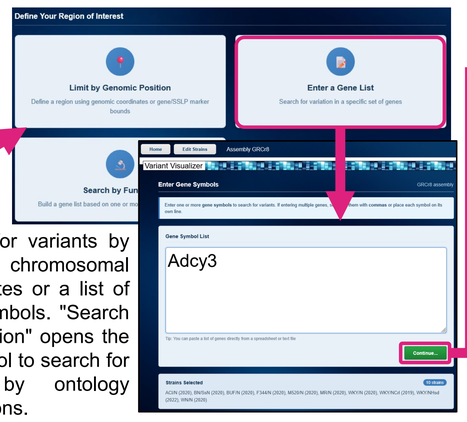


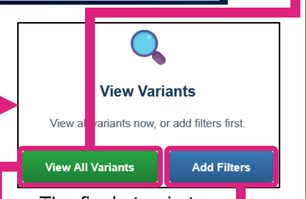


Start with any option on Variant Visualizer's redesigned front page. Choosing "Select Strains" for rat GRCr8 opens the streamlined strain selector page. At the top of the page, HRDP and HS Founder strain groups provide one-click selection of strains in those categories. Below these, samples are listed by strain. Rows, closed by default, open when a group is selected.

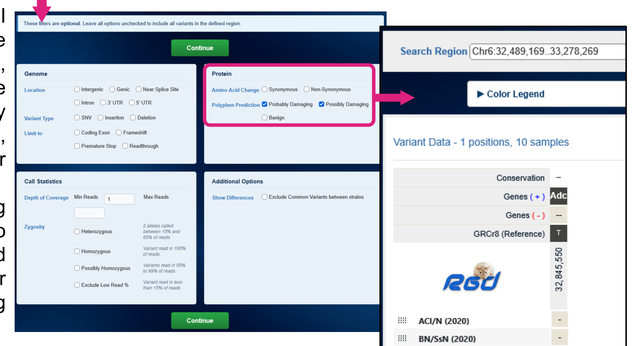


Search for variants by entering chromosomal coordinates or a list of gene symbols. "Search by Function" opens the OLGA tool to search for genes by ontology annotations.



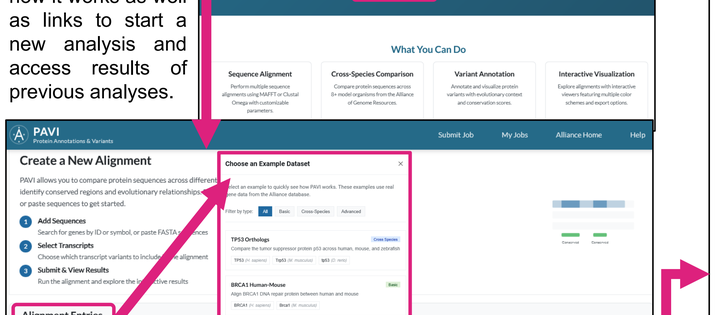


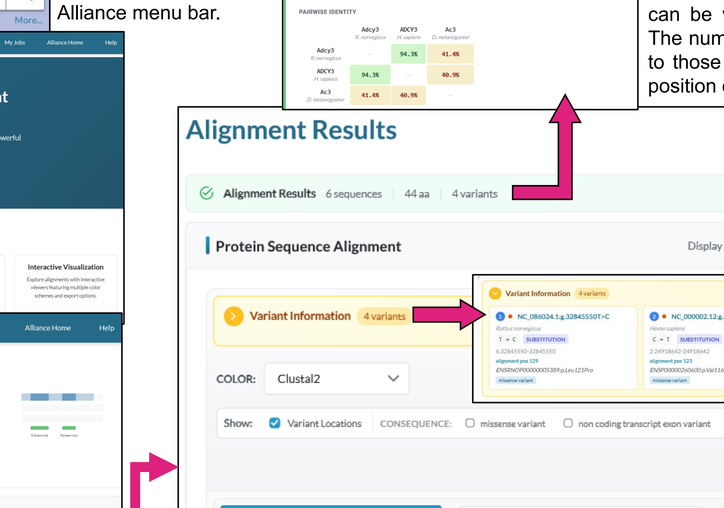
The final step is to choose to view all variants in the region or gene list, or limit the selection by variant attributes, call statistics or consequence, such as limiting the selection to variants predicted to be probably or possibly damaging by Polyphen.



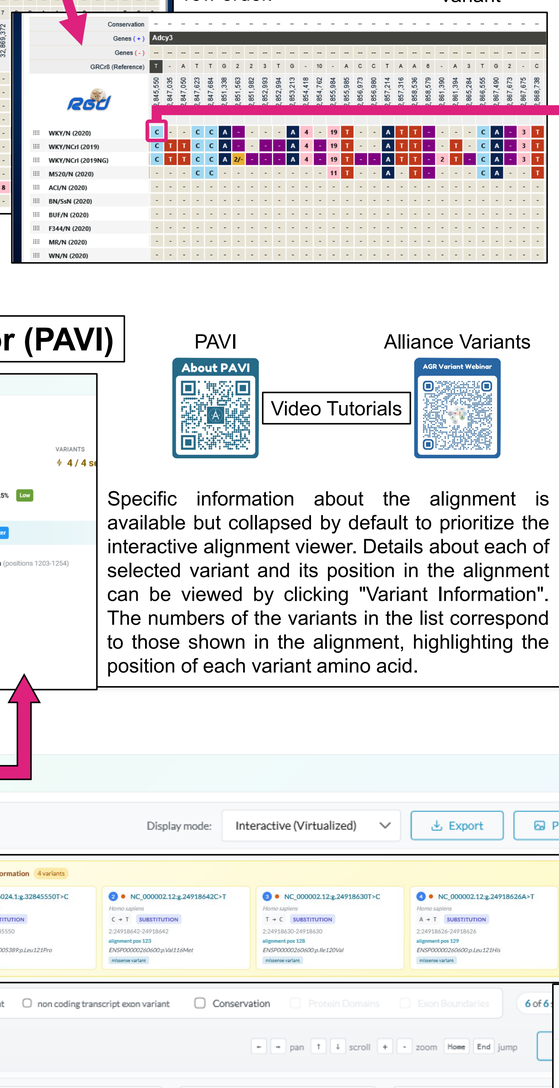
New functionality in the variant viewer makes it easy to adjust the display by zooming in or out using the options at the top of the window. To the left of each sample name, a "handle" allows the user to click and drag the rows to reorder them in the display.

A new "color legend" provides information about the meaning of the colored variant boxes.

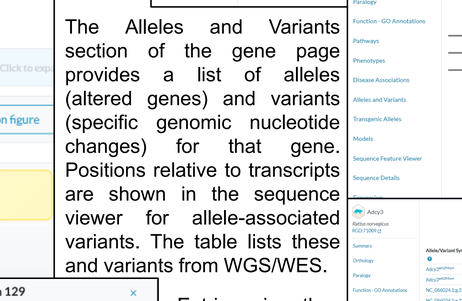


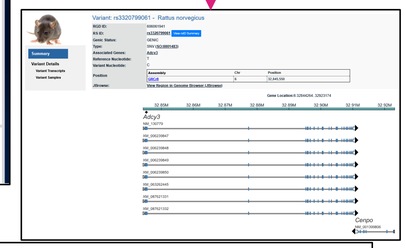


Buttons above the variant display link to an overview of all the variants in that region, group the rows in the display by similarities in variant positions and sequence, and reset the display to the default row order.



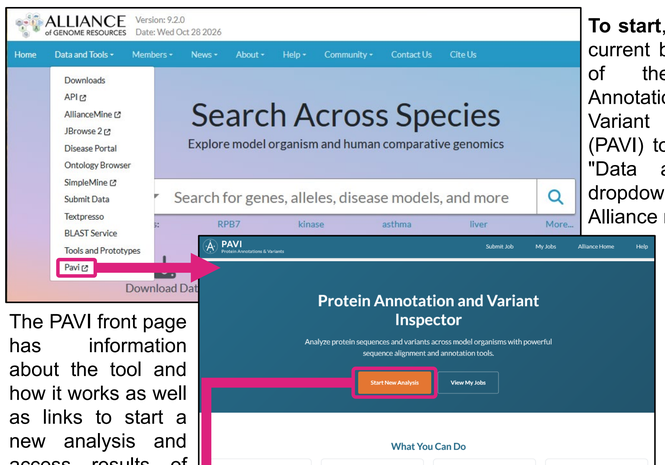
Click a variant box in the display to see a popup with data about that variant





The Variant Details popup links to the corresponding RGD report page, which provides an overview of what is known about that variant including the list of strains where that variant is found.

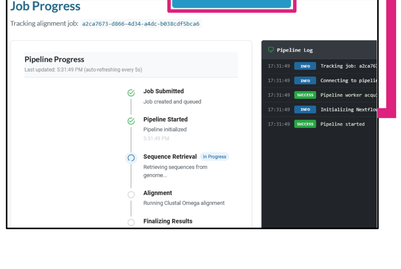
Alliance of Genome Resources: Protein Annotation & Variant Inspector (PAVI)



To start, access the current beta version of the Protein Annotation and Variant Inspector (PAVI) tool from the "Data and Tools" dropdown in the Alliance menu bar.

The PAVI front page has information about the tool and how it works as well as links to start a new analysis and access results of previous analyses.

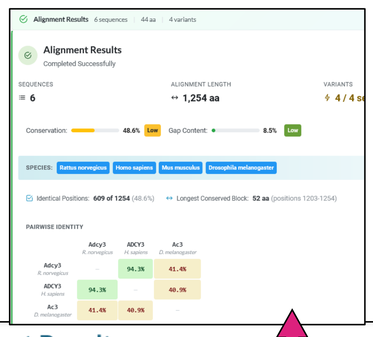
Example analyses are available to demonstrate how the tool works. To begin a new analysis, select one or more genes from any of the Alliance species. For each gene select at least one transcript. If desired, enter one or more variant identifiers for each.



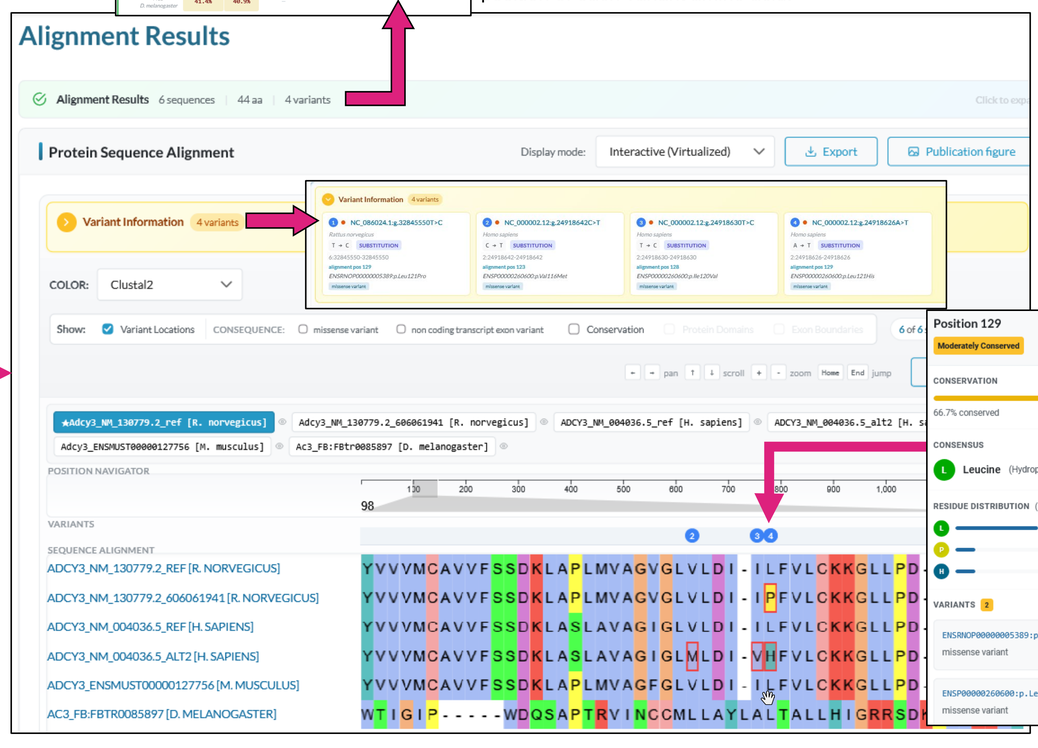
PAVI

Video Tutorials

Alliance Variants



Specific information about the alignment is available but collapsed by default to prioritize the interactive alignment viewer. Details about each of selected variant and its position in the alignment can be viewed by clicking "Variant Information". The numbers of the variants in the list correspond to those shown in the alignment, highlighting the position of each variant amino acid.

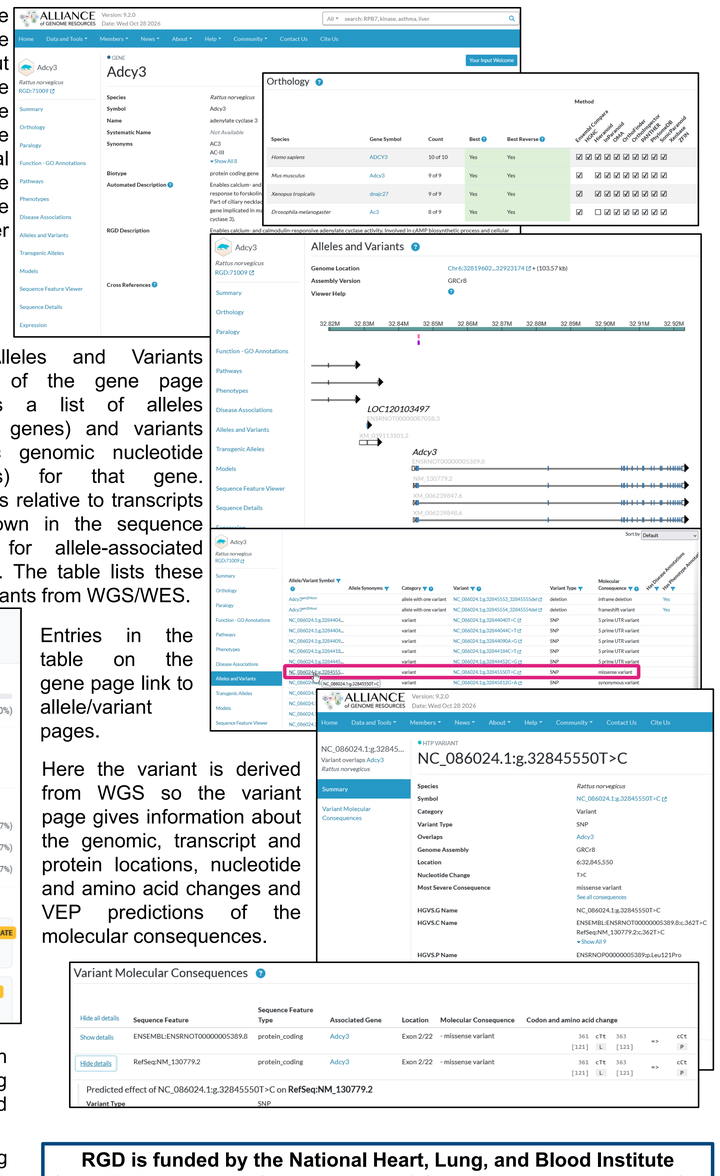


When the job is submitted a progress screen is shown. The tool uses the start and stop positions of the exons for each transcript to create the transcript sequence from the genome assembly sequence and translates it into the corresponding protein. If variants have been selected, the transcript sequence is altered based on the variant position and sequence and translated into an altered protein sequence. The reference proteins and any variant proteins are then aligned.

In the interactive protein alignment view, a gray "POSITION NAVIGATOR" slider shows the portion of the sequence being displayed and can be dragged left or right to move up and down the alignment. Variant amino acids are labeled with numbers and outlined in red. The default color scheme, Clustal2, determines color by the physicochemical properties and conservation of the amino acids. Click on a column in the alignment to see information about the amino acids at that position in the aligned sequences.

Alliance of Genome Resources: Gene/Variant Pages

All Alliance gene pages share the same layout regardless of the species. Here the rat Adcy3 gene page gives general information at the top and lists the orthologs in other Alliance species.



The Alleles and Variants section of the gene page provides a list of alleles (altered genes) and variants (specific genomic nucleotide changes) for that gene. Positions relative to transcripts are shown in the sequence viewer for allele-associated variants. The table lists these and variants from WGS/WES.

Entries in the table on the gene page link to allele/variant pages.

Here the variant is derived from WGS so the variant page gives information about the genomic, transcript and protein locations, nucleotide and amino acid changes and VEP predictions of the molecular consequences.

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